

Wireless AC Automation System

This automation system enables wireless control of a split AC unit in a remote room using a mesh network between two ESP32 nodes. The Controller Node connects to an existing network switch via an Ethernet module (W5500 / ENC28J60), while the Sensor Node in the target room handles IR transmission and environmental sensing.

The solution is flexible, infrastructure-independent, and scalable — each Controller Node can manage up to two Sensor Nodes. A browser-based control panel allows full AC management and real-time environmental monitoring from any device on the network.

System Architecture

The system consists of two ESP32 firmware roles communicating over a self-forming painlessMesh WiFi network — no separate router or access point is required between nodes.

Node	Location	Role
Controller Node	Network Cabinet	Hosts web dashboard · Ethernet to network switch · Mesh coordinator
Sensor Node	Target Room	DHT22 sensing · IR AC transmission · Mesh client

Key Components

- Sensor Node (ESP32)** — DHT22 temperature & humidity sensor; IR LED transmitter for AC command delivery; periodic heartbeat broadcast via mesh.
- Controller Node (ESP32)** — W5500/ENC28J60 Ethernet module for wired network connectivity; FreeRTOS dual-core architecture (Ethernet: Core 0, Mesh: Core 1); NVS storage for registered node IDs and locations.
- W5500 / ENC28J60 Ethernet Module** — SPI-based wired Ethernet for reliable communication with the network switch.

System Functionality

Wireless Control	Sensor Node receives commands from Controller via mesh and translates them into IR signals for the AC unit.
Bidirectional Comms	Ethernet (W5500/ENC28J60) provides two-way data exchange between the Controller and the network. Mesh provides two-way comms between nodes.
Remote Management	AC unit controlled from any browser on the local network — no internet or cloud required.
IR Translation	ESP32 converts JSON AC commands into precise IR signals using the IRremoteESP8266 library (Coolix/HVAC protocol).
Node Registration	Each Sensor Node is identified by its unique ESP32 Chip ID. Operators assign a location label via the web dashboard; data persists in NVS.

Heartbeat Monitoring	Controller tracks each node's last-seen timestamp. Nodes not heard from within 5 s are flagged OFFLINE on the dashboard.
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Hardware & Software Requirements

- 2 × ESP32 Development Board + USB cable
- DHT22 Temperature & Humidity Sensor
- IR Transmitter (TSAL6100) + AO3400A MOSFET driver
- W5500 or ENC28J60 Ethernet Module
- 10 μ F, 100 nF bypass capacitors
- 1 k Ω , 10 k Ω , 100 k Ω resistors
- MEAN WELL IRM-03-3.3 / IRM-02-3.3 AC-DC module (85–264 VAC in, 3.3 VDC out)
- Arduino IDE (free) with ESP32 board support
- Libraries: painlessMesh, Arduino_JSON, IRremoteESP8266, DHT (Adafruit), EthernetENC

Implementation Steps

- Decode AC remote control IR signals using ESP32 + IR receiver module (TSOP38238).
- Implement DHT22 sensor reading and IR transmission on the Sensor Node ESP32.
- Establish painlessMesh network between Controller and Sensor Node ESP32s.
- Connect Controller Node to network switch via ENC28J60/W5500 Ethernet module.
- Control the AC from any browser using the Controller Node's static IP address.
- After prototype validation, design and fabricate a custom PCB integrating all components.

Scalability & Deployment

Each Controller Node supports up to **2 Sensor Nodes**, making it straightforward to scale across multiple rooms from a central cabinet. A Grafana dashboard can be integrated for centralized monitoring and historical data logging across all nodes.

Controller #1	Room A + Room B
Controller #2	Room C + Room D
Controller #N	Room ... + Room ...

Planned Features (Roadmap)

- Over-the-Air (OTA) firmware updates for both nodes
- CT sensor integration for AC current & power consumption monitoring
- Grafana / InfluxDB dashboard integration

- HTTPS and basic authentication for the web panel